INCH-POUND

MIL-PRF-1/292E
14 June 2007
SUPERSEDING
MIL-E-1/292D
23 February 1987

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 5829WA

Inactive for new design
after 7 March 1997.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and MIL-PRF-1.

DESCRIPTION: Double diode, subminiature, 400 MHz.

Outline - - - 2-5 (EIA).
Base - - - Pinch press (7 leads in line).
Envelope - - - T2X3.
Cathode - - - Coated unipotential.
Base connections:

Pin No.	- - -	1	2	3	4	5	6	7
Element	- - -	2a	2k	h	sd	h	1a	1k

ABSOLUTE RATINGS:

Parameter:	Ef	Epp/a	epx	Ehk	R1	C1	Io/a	ib/a	isurge/a	TE	Alt
Unit:	V	V ac	v	v	Ohms	μF	mA dc	mA	mA	°C	ft
Maximum:	6.6	130	360 (see note 1)	360	- - -	- - -	5.5	33	175	220	See note 8
Minimum:	6.0	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -	- - -
Test conditions:	6.3	117	- - -	0	14,000	8	- - -	- - -	- - -	- - -	- - -

GENERAL:

Qualification: Not required.

Reliable tube.

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TABLE I. Requirement or test.

MIL-STD-1311 method	Requirement or test	Notes	Conditions	Acceptance level (See note 11)	Symbol	Limits		Unit
						Min	Max	
	First article inspection							
2220	Resonant frequency			- - -	Length	13.5	- - -	cm
	Conformance inspection, part 1							
1201	Short and discontinuity detection			0.4	- - -	- - -	- - -	- - -
1301	Heater current			0.65	If	138	162	mA
1231	Emission	2	Es = 6.5 V dc	0.65	Is	15	- - -	mA dc
1353	Operation of rectifiers	3, 4		0.65	Io	9.0	- - -	mA dc
1336	Heater-cathode leakage	2		0.65	Ihk	- - -	10	μA dc
	Conformance inspection, part 2							
1211	Insulation of electrodes	2		2.5	R	- - -	- - -	- - -
1256	Electrode current (anode)	2	Ebb = 0; Rp = 40,000 ohms	2.5	Ib	2	20	μA dc
1256	Electrode current (anode) (difference between sections)			2.5	Ib	- - -	5	μA dc
1331	Direct-interelectrode capacitance		No shield	6.5	C			
			1a to 2a			0.06	0.12	pF
			1a to h+1k+sd			1.9	3.5	pF
			2a to h+2K+sd			1.7	3.3	pF
			1k to h+1a+sd			2.4	4.2	pF
			2k to h+2a+sd			2.8	4.6	pF
			1k to h			1.1	2.2	pF
			2k to h			1.3	2.5	pF
1116	Lead fatigue			2.5	- - -	- - -	- - -	- - -
2126	Envelope strain			2.5	- - -	- - -	- - -	- - -
1041	Shock		450 G	15	- - -	- - -	- - -	- - -
1031	Variable-frequency vibration	5	No voltages	10	- - -	- - -	- - -	- - -

See notes at end of table.

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TABLE I. Requirement or test - Continued.

MIL-STD-1311 method	Requirement or test	Notes	Conditions	Acceptance level (See note 11)	Symbol	Limits		Unit
						Min	Max	
	Conformance inspection, part 2 - Continued							
1031	Vibration-fatigue	5		6.5	- - -	- - -	- - -	- - -
---	Shock, variable-frequency vibration, and vibration-fatigue-test end points:							
1336	Heater-cathode leakage			- - -	Ihk	- - -	15	μA dc
1353	Operation of rectifiers			- - -	Io	7.0	- - -	mA dc
1105	Permanence of marking			- - -	- - -	- - -	- - -	- - -

MIL-STD-1311 method	Requirement or test	Notes	Conditions	Symbol	Limits		Unit
					Min	Max	
	Conformance inspection, part 3						
1506	Heater-cycling life-test		Ef = 7.5 V; Ehk = 140 V ac; Eb = 0; 1 min on, 4 min off	- - -	- - -	- - -	- - -
- - -	Heater-cycling life-test end point:						
1336	Heater-cathode leakage			Ihk	- - -	20	μA dc
1501	Intermittent life	6 ,7,10	TE = 220°C (min); Ehk = 117 V ac; Group E	- - -	- - -	- - -	- - -
- - -	Intermittent life-test end points (500 hours):						
1301	Heater current			If	135	165	mA
1353	Operation of rectifiers			Io	7.5	- - -	mA dc
1353	Change in operation of rectifiers of individual tubes			ΔIo t	- - -	15	%

See notes at end of table.

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TABLE I. Requirement or test - Continued.

MIL-STD-1311 method	Requirement or test	Notes	Conditions	Symbol	Limits		Unit
					Min	Max	
	Conformance inspection, part 3 - Continued						
1336	Heater-cathode leakage			Ihk	- - -	20	μA dc
1211	Insulation of electrodes			R	50	- - -	MegΩ
---	Intermittent life-test end points (1,000 hours):						
1301	Heater current			If	135	165	mA
1353	Operation of rectifiers			Io	7.5	- - -	mA dc
1353	Change in operation of rectifiers of individual tubes			ΔIo t	- - -	15	%
1336	Heater-cathode leakage			Ihk	---	20	μA dc
1211	Insulation of electrodes			R	25	- - -	MegΩ

NOTES:

1. The maximum voltage appearing between any pair of pins shall be no greater than the peak inverse anode voltage rating.
2. Test each unit separately.
3. In a full-wave circuit, adjust Zp/a so that Io = 10 mA dc with a TUT having an Etd of 5.5 V dc at 15 mA dc per anode. The minimum peak anode current shall be 25 mA per anode. Tap and reject for recurring arcs or sputters.
4. This test shall be performed at the conclusion of the holding period.
5. This test shall be conducted on the initial lot and thereafter on a lot approximately every 12 months. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot has passed. When one lot has passed, the 12-month rule shall apply.
6. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (MIL-STD-1311, method 1226), will be satisfied if a TUT having bogey Ib (± 5 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.
7. In a full-wave life-test circuit, the values of R1 and C1 shall be considered as approximate and shall be adjusted initially so that Io is equal to or greater than 10 mA dc with ib equal to or greater than 25 mA.
8. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.
9. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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NOTES - Continued:

10. The life test sample shall consist of 20 tubes per lot and not one tube failure shall be permitted. In the event of rejection of the first sample, a second sample of 40 tubes shall be selected from the lot. Acceptance shall be based on the second sample and shall not permit any tube failures.

11. This specification sheet utilizes an accept on zero defect sampling plan in accordance with MIL-PRF-1, table III.

Referenced documents: In addition to MIL-PRF-1, this document references MIL-STD-1311.

Changes from the previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5960-2007-024)

Review activities:
Army - AR, MI
Navy - AS, CG, MC, OS
Air Force - 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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